PCA CODE

```
from sklearn.decomposition import PCA
# Apply PCA to reduce dimensionality (fit only on training data)
pca = PCA(n_components=2)
X_train = pca.fit_transform(X_train)
X_test = pca.transform(X_test)
```

LDA CODE

```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
lda = LinearDiscriminantAnalysis(n_components=2)
X_train = lda.fit_transform(X_train, y_train)   # needs labels
X_test = lda.transform(X_test)
```

Kernel PCA

```
from sklearn.decomposition import KernelPCA
kpca = KernelPCA(n_components=2, kernel='rbf', gamma=15)
X_train = kpca.fit_transform(X_train)
X_test = kpca.transform(X_test)
```

The below code for Reference for Kernal PCA

```
from sklearn.decomposition import KernelPCA
kpca = KernelPCA(
    n_components=2,
    kernel='rbf',       # 'linear' | 'poly' | 'rbf' | 'sigmoid' | 'cosine' | 'precomputed'
    gamma=15,           # used by 'rbf', 'poly', 'sigmoid'
    degree=3,           # used by 'poly'
    coef0=1             # used by 'poly', 'sigmoid'
)
X_train = kpca.fit_transform(X_train)
```

```
X_test = kpca.transform(X_test)
```